

Homework Set 3, CPSC 6430/4430

LastName, FirstName

Due 03/31/2026, 11:59PM EST

PCA

Please refer to Jupyter Notebook.

Naive Bayes

The table below provides a training dataset containing sixteen observations. Now please determine by Naive Bayes whether the new coming data is Male (M) or Female (F) given ‘Basketball, 6 feet (height), 150lbs, 4 inches hair, 10 inches footsize and Drinking’.

Sports	Height (feet)	Weight (lbs)	Hair (inches)	Footsize (inches)	Drinking	Class
Basketball	6	145	6	11	No	F
Tennis	5.6	150	8	11	Yes	M
Basketball	6.4	170	3	13	Yes	M
Basketball	6.2	150	7	12	No	M
Basketball	6	140	6	11	Yes	F
Tennis	6.4	130	7	12	No	F
Soccer	5.8	140	4	12	Yes	M
Soccer	6.2	130	7	12	No	F
Soccer	6	150	4	10	Yes	M